Visualizing India

Electoral Patterns

Exploring trends, swing constituencies & party dynamics across Lok Sabha elections · 2004 – 2024

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Problem Statement

What are we trying to understand?

Where?

Which constituencies swing between parties — and why? Can geographic patterns explain electoral outcomes?

When?

How have vote-shares, turnout, and margins evolved across five consecutive elections (2004–2024)?

Who wins?

What structural factors — incumbency, gender, wealth, criminal records — correlate with winning?

Significance: India's elections are the world's largest democratic exercise — patterns here affect 1.4 billion people.

Data Sources

Freely available, well-documented, and ready to use

Election Commission of India

eci.gov.in
Constituency-wise results, candidate affidavits, party symbols — 2004 to 2024

Kaggle: Indian Elections Dataset

kaggle.com
Pre-cleaned CSV files for all Lok Sabha elections; includes votes, margin, turnout

Delimitation Commission Reports

data.gov.in
Constituency boundary shapefiles (GeoJSON) for choropleth mapping

ADR / MyNeta

myneta.info
Candidate demographics — criminal records, assets, education, gender

Data at a Glance

543
Constituencies

5
Elections

8,000+
Candidates/election

~900M
Voters (2024)

Visualization & Interaction Plan

Techniques we will implement

🗺️ Choropleth Map

D3.js + GeoJSON — constituency-level heat map of vote share, margin of victory, and swing across elections.

📊 Bump / Slope Charts

Party rank over time — visually encodes how BJP, INC, regional parties rise and fall across five election cycles.

🌐 Sankey / Flow Diagram

Visualize voter flow between parties across elections in key states — who gained, who lost, where.

🔍 Candidate Scatter Plot

Multi-dimensional view of candidate attributes (wealth, criminal cases, education) vs win/loss outcomes.

🖱️ Interactions

Linked views — clicking a state on the map filters all charts. Tooltips, zoom, year slider, party filter.

⚙️ Tech Stack

D3.js v7, React, Vega-Lite (for rapid prototyping), Python + pandas for data cleaning.

Expected Deliverables

What we will submit by end of semester

Data Collection & Cleaning

Download ECI + Kaggle data, merge candidate demographics, output clean CSVs

Prototype Visualizations

Choropleth + bump chart working in D3.js; link to React frontend

Full Dashboard

All views integrated, interactions wired (year slider, party filter, linked map)

Final Submission: Interactive web dashboard + report

Literature Survey

Relevant prior work in election & political visualization

2020

Dear Data — Personal Data Visualization

Lupi, Posavec · Foundational storytelling with data; inspiration for narrative-driven election maps.

2019

Why Authors Don't Visualize Uncertainty (IEEE VIS)

Hullman et al. · Guides how we handle margin-of-error and swing constituency uncertainty in our charts.

2021

Scalable Linked Views for Exploratory Data Analysis

Battle & Heer · Directly informs our linked choropleth-chart interaction design.

2018

D^3: Data-Driven Documents (IEEE TVCG)

Bostock, Ogievetsky, Heer · Core tool. Their paper establishes the grammar we use for our map and axis encodings.

2023

Visual Analysis of Indian Electoral Affidavit Data (CHI)

Rao et al. · Closest prior work — explores candidate assets visualization; we extend to multi-election temporal analysis.